Princess Nora Bint Abdulrahman University

Faculty of Computer and Information Sciences

Department of Information Technology

Course: Network Fundamentals

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Section: O42

Group number: 2

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Firewall blocking	Reuf
MAC/Port Security	Layan,Reuf,Ajwan,Ghaleh
Report writing	Layan, Reuf, Ajwan, Ghaleh

2. Introduction

How did we make the network?
We set up a LAN network connecting devices in the same area using a router and three RJ-45 cables between the router and the laptops. Each laptop was assigned a static IP within the same subnet. We performed connectivity tests like pinging, sharing folders, using LAN communication software such as BeeBEEP and Softros, and enabling Remote Desktop between the devices.

The type of physical topology used

Star Topology
main premise of a star topology is that devices are individually connected via a central networking device such as a switch or hub. This topology is the most commonly found today because of its reliability and scalability — despite the cost.
Because more cabling and the purchase of dedicated networking equipment is required for this topology, it is more expensive than any of the other topologies. However, despite the added cost, this does provide some significant advantages. For example, this topology is much more scalable in nature, which means that it is very easy to add more devices as the demand for the network increases. [1]

Tasks Implementation

4.1 Ping Test

<u>Purpose: To verify network connectivity between three devices.</u>

1) Turn on all sharing settings from change advanced sharing settings.
2) Change properties for local connection from change adapter settings to manually write IP addresses for all devices.
3) From CMD window test the ping for all devices by the line (ping IP address). [2]

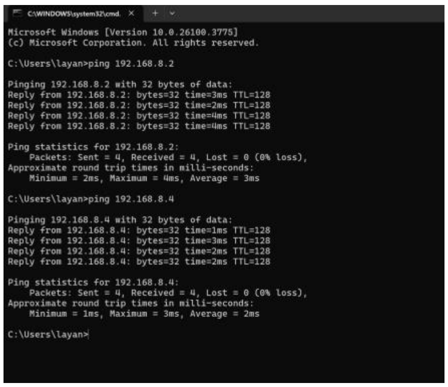


Figure 1:Device 1 ping test

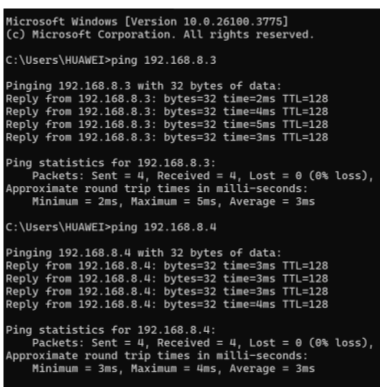


Figure 2:Device2 ping test

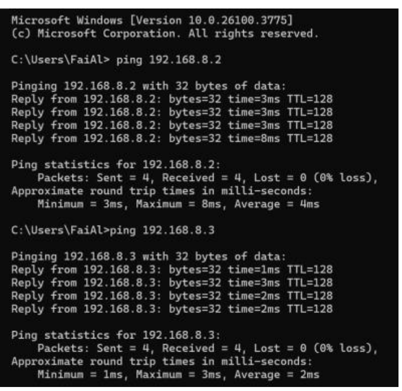


Figure 3: Device3 ping test

4.2 Remote Desktop Control

<u>**Purpose:** To remotely access and control one device from another.</u>

Install & launch TeamViewer in all devices. [3]

1) Each device will have an ID and a Password (credentials used to connect).
2) Device 1 and Device 2 entered the ID and password showed in Device 3.

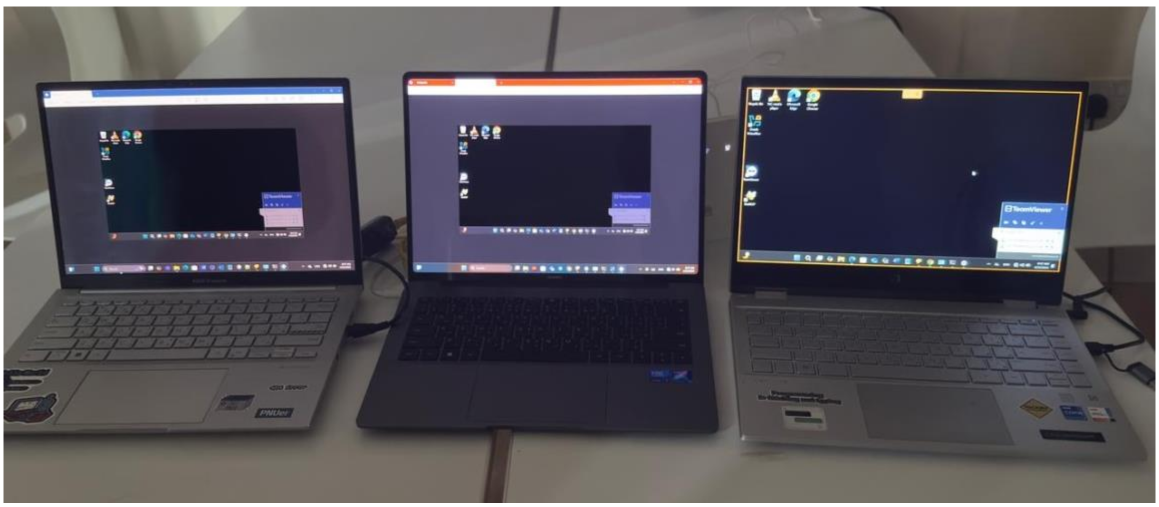


Figure 4: Device1 and Device2 are controlling Device3

4.3 Firewall Configuration

Purpose: To block a specific LAN chat application (BeeBEEP) using Windows Firewall. create inbound & outbound Rules from system security>defender firewall settings.

Choose **BeeBEEP** and block the connection.

BeeBEEP Before Block

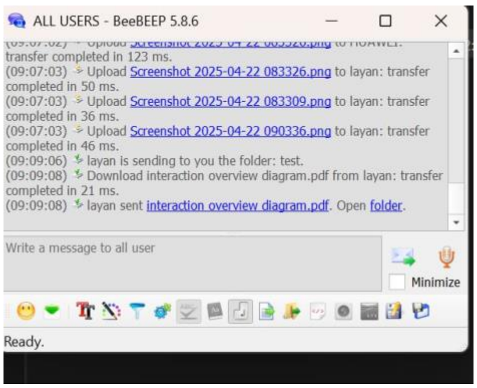


Figure 5: Device1 sent Folder to All Devices

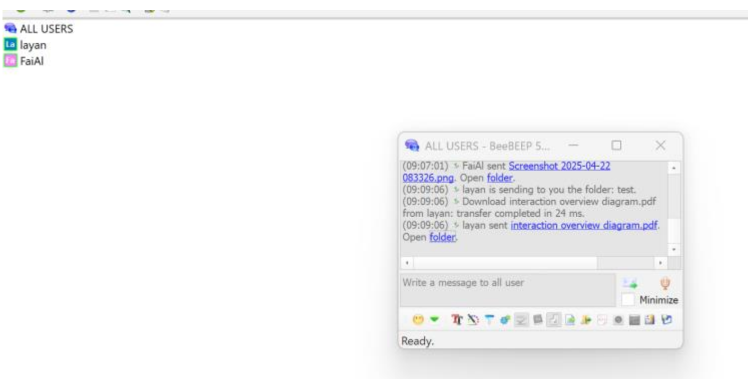


Figure 6: Device2 Received file

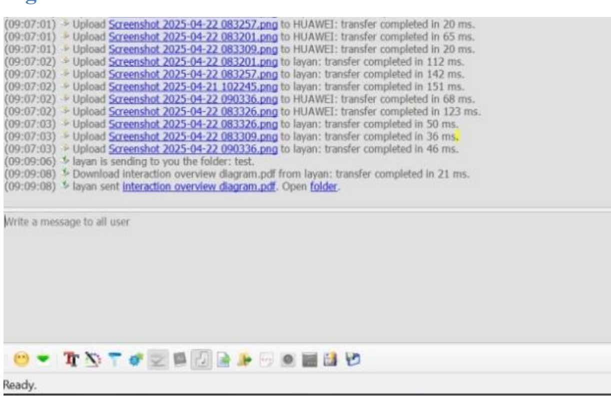


Figure 7: Device3 Received file

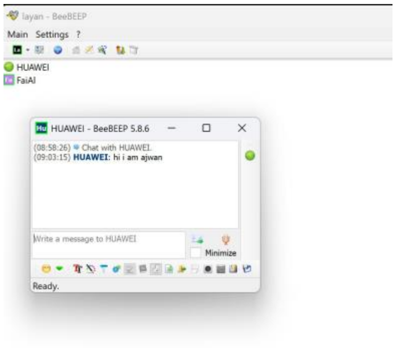


Figure 8: Device2 chat with Device1

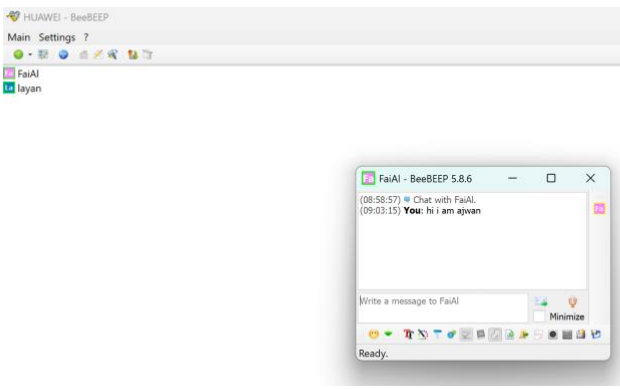


Figure 9: Device2 chat with Device3

Firewall configuration process

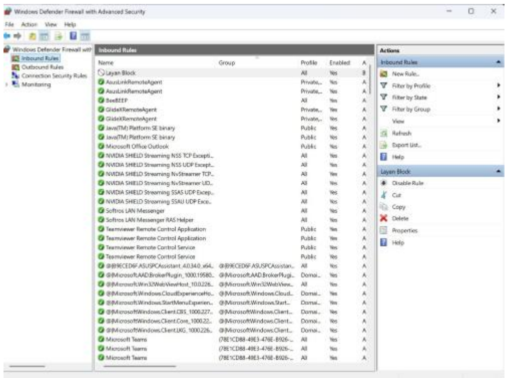


Figure 10: Device1 Blocked chat program

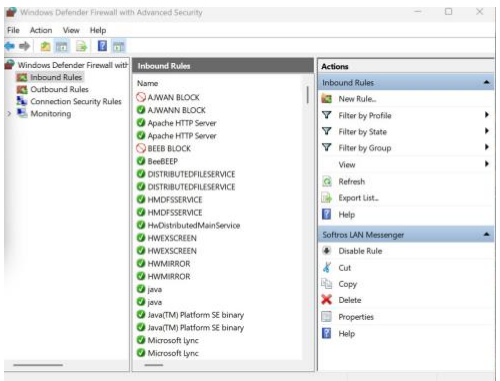
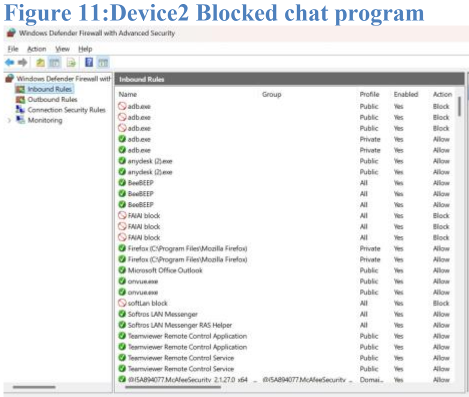


Figure 11:Device2 Blocked chat program



BeeBEEP is Blocked

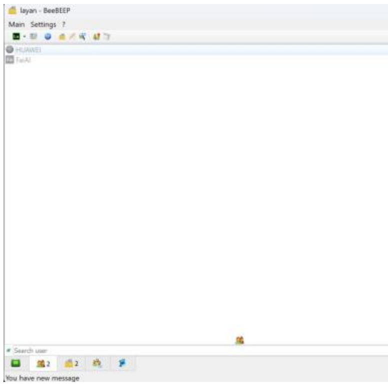
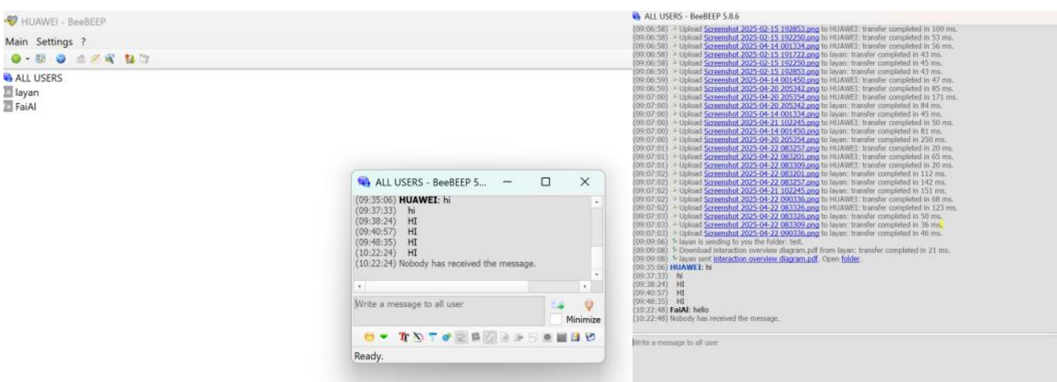


Figure 12: All devices went offline-Device1 POV

Figure 13: Messages is not received



4.4 Basic Network Security

Purpose: To restrict network access to authorized devices using MAC filtering.

1) Log into Router Admin panel.
2) Enable MAC filtering from Security settings.
3) Add the MAC addresses of the devices to limit their access.

Screenshots:

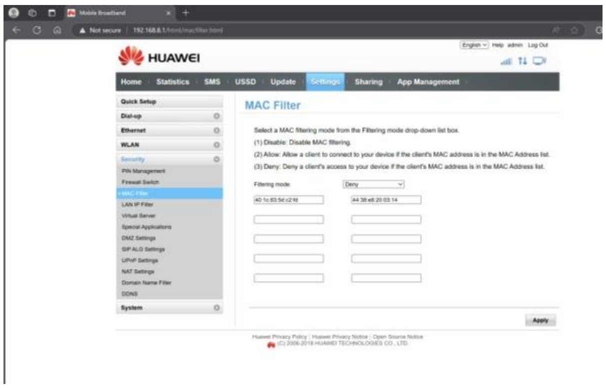


Figure 14: Device 1 added MAC addresses to limit access

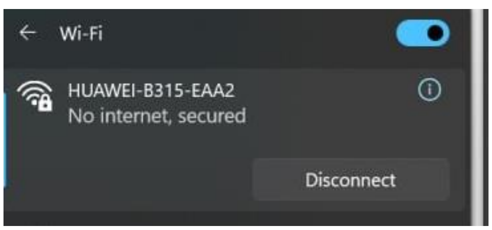


Figure 15: Device2 is not able to connect

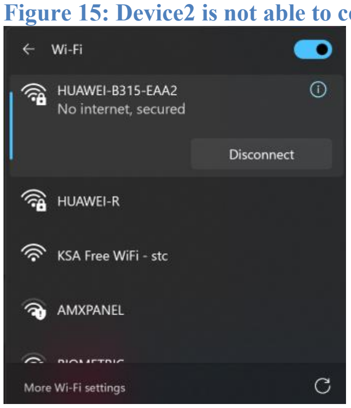


Figure 16: Device3 is not able to connect

4. Problems and Solutions

Problem Encountered	How it was Solved
The router only allowed connection to two devices	We replaced the router with another one that supports 3 device connections
The devices were not connecting properly using the LAN Messenger application	We switched to another LAN communication software called BeeBEEP, which worked successfully
The firewall block was not working as expected for the LAN communication	We configured both inbound and outbound rules for the LAN messaging application in the Windows Firewall settings.

5. References

مراجع

[1] HammadM ،"https://medium.com/@hammaadm/tryhackme-intro-to-lan-11070d9d0253،" 4 December 2023 .[متصل] .

[2] A. Sentika ،"https://www.hostinger.com/tutorials/linux-ping-command-with-examples،" 1 March 2024 .[متصل] .

[3] J. Pot ،"https://me.pcmag.com/en/system-utilities/18927/the-best-remote-access-software-for-2023،" 21 April 2015 .[متصل] .